

Qatar-1b

R-0399

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180911 · created 2026-07-19T03:47:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458372.6586 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.038 +/- 0.056
Transit depth (Rp/Rs)^2	0.14 +/- 0.42 [%]
Orbital Inclination (inc)	79.75 +/- 2.89
Airmass coefficient 1 (a1)	5.7 +/- 3.06
Airmass coefficient 2 (a2)	-1.45 +/- 1.74
Scatter in the residuals of the lightcurve fit is	53.44 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.5
Transit Duration (day)	0.018 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.038 ± 0.056 vs 0.1463 (deviation 1.93σ)
Duration fit vs published	0.018 d vs 0.0692 d (deviation 2.13σ)
Mid-time O-C	-1.9 min
Depth SNR	0.7
Residual scatter	53.44 %

Light curve

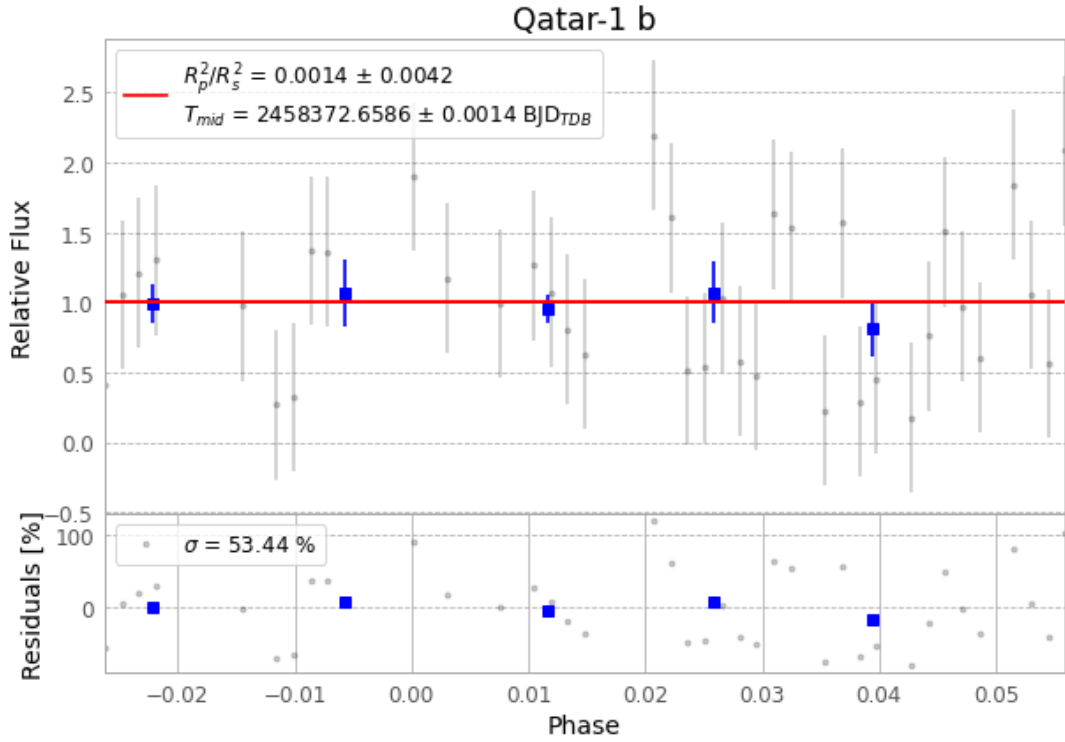


Figure 1 — FinalLightCurve_Qatar-1 b_2018-09-10.png (SHA-256 f061d85a20af6021...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [108, 368], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 15a1c322dc11621f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

57 FITS frames (37 MB), Qatar-1180911025112.FITS → Qatar-1180911053912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180911.log (b02c29e93444...), FinalLightCurve_Qatar-1 b_2018-09-10.png (f061d85a20af...), FinalLightCurve_Qatar-1 b_2018-09-10.csv (c66b60f53be9...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 03:47Z.